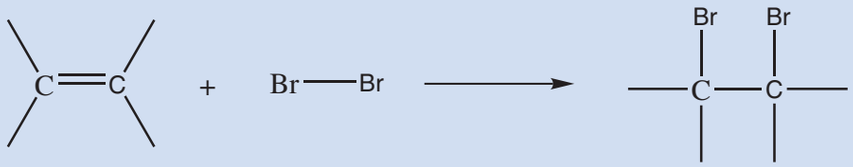
6. Ignition test

Aromatic compounds burn with a strong sooty yellow flame because of the high carbon–hydrogen ratio. Aliphatic compounds burn with non-sooty flame.

7.Test with bromine water:

In this test, the orange-red colour of bromine solution disappears when it is added to an unsaturated organic compound.

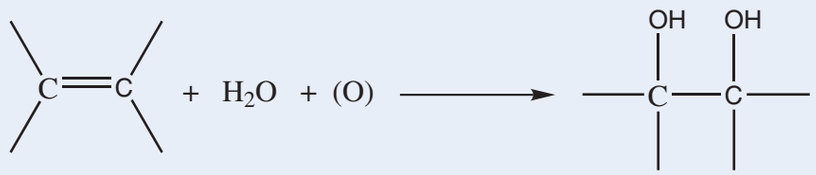


unsaturated compound — Orange-yellow — Colour less

8. Test with $KMnO_4$ (Baeyer's Test)

In this test, pink colour of $KMnO_4$ disappears, when alkaline $KMnO_4$ is added to an unsaturated hydrocarbon. The disappearance of pink colour may take place with or without the formation of brown precipitate of MnO_2.

$$2KMnO_4 + H_2O \longrightarrow 2KOH + 2MnO_2 + 3(O)$$



unsaturated compound — Colour less

9. Neutral FeCl3 test:

Phenol reacts with ferric ions to form violet coloured complex.

Aqueous solution Naphthols do not give any characteristic colour with neutral ferric chloride. But alcoholic solution of α and β naphtholsgiveblue-violet and green colouration respectively due to the formation of binaphthols.

organic analysis.indd 261 3/29/2019 6:48:21 PM